🌍 Interactive 3D Earth Portfolio — Project Plan

Overview

A stunning, interactive 3D globe portfolio where projects are pinned to geographic locations. Users can rotate the Earth, click on project markers, and filter by continent.

🎯 Core Features (from reference image)

1. 3D Earth Globe

- **Technology**: Three.js with custom shaders
- **Visuals**: Realistic Earth texture with night lights, atmosphere glow, starfield background
- **Interaction**: Mouse/touch drag to rotate, scroll to zoom, auto-rotation when idle
- **Loading**: Animated progress ring (0% → 100%) while textures load

2. Project Pins

- **Placement**: Projects pinned to real geographic coordinates
- **Visuals**: Glowing markers with connection arcs between related projects
- **Labels**: Project name + location shown on hover/click
- **Animation**: Markers pulse with a subtle glow effect

3. Continental Filter Bar

- **Regions**: All Earth | North America | Europe | Africa | Asia | Australia | South America | Arctic
- **Behavior**: Clicking a region rotates globe to focus on that area
- **Highlight**: Active region gets highlighted styling

4. UI/UX Details

- **Theme**: Dark mode (deep space black background)
- **Typography**: Clean sans-serif, white/light text
- **Header**: Name + "Interactive 3D Earth Portfolio" tagline
- **Audio Toggle**: Sound effects on/off button
- **Contact CTA**: "say hello →" link
- **Instructions**: "Click on pinned locations to explore projects. Drag to rotate."

Tech Stack

Layer	Technology
3D Engine	Three.js (r160+)
Shaders	Custom GLSL (atmosphere, earth surface, glow)
UI Framework	Vanilla HTML/CSS/JS (or React if preferred)
Animation	GSAP for UI transitions
Textures	NASA Blue Marble (day/night), custom bump/ specular maps

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Table 1 – continued

Layer	Technology
Icons	Lucide or custom SVG
Fonts	Inter or Geist

📁 File Structure

```
3d-earth-portfolio/
├── index.html
├── css/
│   ├── main.css
│   ├── globe.css
│   └── ui.css
├── js/
│   ├── main.js
│   ├── globe.js          # Three.js scene setup
│   ├── earth.js          # Earth mesh & shaders
│   ├── markers.js        # Project pin system
│   ├── interactions.js   # Mouse/touch controls
│   └── ui.js             # Filter bar, modals, loading
├── assets/
│   ├── textures/
│   │   ├── earth-day.jpg
│   │   ├── earth-night.jpg
│   │   ├── earth-bump.jpg
│   │   ├── earth-specular.jpg
│   │   └── stars.jpg
│   └── projects.json     # Your project data
└── README.md
```

🎨 Visual Design System

Colors

- Background: `#000000` (pure black) → `#0a0a0f` (deep space)
- Earth: Realistic satellite imagery
- Atmosphere: `#4a90d9` (soft blue glow)
- Markers: `#00ff88` (neon green) with `#ffffff` core
- Text Primary: `#ffffff`
- Text Secondary: `#a0a0a0`
- Accent: `#00d4ff` (cyan for interactive elements)

Typography

- Name: 24px, weight 700, letter-spacing 0.05em
- Tagline: 12px, weight 400, uppercase, opacity 0.6

- Project Labels: 14px, weight 500
- Filter Buttons: 13px, weight 500, uppercase

Spacing

- Globe centered, ~60vh height
- Filter bar: bottom-center, 16px from bottom
- Side panels: 24px from edges

🚀 Implementation Phases

Phase 1: Core Globe (Week 1)

- Three.js scene with camera, renderer
- Earth sphere with day texture
- Atmosphere shader (Fresnel glow)
- Starfield background
- OrbitControls for rotation/zoom
- Loading screen with % progress

Phase 2: Project System (Week 2)

- Marker placement from lat/lng
- Marker glow/pulse animation
- Project data JSON structure
- Hover/click interactions
- Connection arcs between projects

Phase 3: UI Polish (Week 3)

- Continental filter bar
- Region-based camera transitions
- Project detail modal/card
- Audio toggle
- Responsive design (mobile touch)

Phase 4: Content & Deploy (Week 4)

- Add your real projects with coordinates
- Custom project screenshots
- Performance optimization
- Deploy to Vercel/Netlify

📝 Project Data Format

```
{
  "projects": [
    {
```

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```
    "id": "dsa-tracker",
    "name": "DSA Tracker Pro",
    "description": "Comprehensive data structures & algorithms progress tracker",
    "location": {
      "name": "Arctic / Greenland",
      "lat": 71.7069,
      "lng": -42.6043
    },
    "continent": "arctic",
    "tech": ["React", "TypeScript", "Node.js"],
    "link": "https://github.com/you/dsa-tracker",
    "image": "assets/projects/dsa-tracker.png"
  },
  {
    "id": "mastermind",
    "name": "MasterMind",
    "description": "AI-powered code review & mentorship platform",
    "location": {
      "name": "Europe",
      "lat": 48.8566,
      "lng": 2.3522
    },
    "continent": "europe",
    "tech": ["Next.js", "OpenAI", "PostgreSQL"],
    "link": "https://github.com/you/mastermind",
    "image": "assets/projects/mastermind.png"
  },
  {
    "id": "mocktest",
    "name": "MockTest365",
    "description": "Daily mock test platform for competitive exams",
    "location": {
      "name": "Africa",
      "lat": -1.2921,
      "lng": 36.8219
    },
    "continent": "africa",
    "tech": ["Vue.js", "Firebase", "Tailwind"],
    "link": "https://github.com/you/mocktest365",
    "image": "assets/projects/mocktest365.png"
  },
  {
    "id": "body-planner",
    "name": "Body Planner",
    "description": "AI fitness & nutrition planning application",
    "location": {
      "name": "North America",
      "lat": 37.7749,
      "lng": -122.4194
    },
    "continent": "north-america",
```

```
      "tech": ["React Native", "TensorFlow.js", "AWS"],
      "link": "https://github.com/you/body-planner",
      "image": "assets/projects/body-planner.png"
    }
  ]
}
```

🎭 Animation Specifications

Element	Animation	Duration	Easing
Globe rotation	Continuous auto-rotate	∞	Linear
Marker pulse	Scale 1.0 → 1.3 → 1.0	2s	Sine.inOut
Marker glow	Opacity 0.4 → 1.0 → 0.4	2s	Sine.inOut
Loading ring	Rotate 360° + scale pulse	1.5s	Linear
Camera transition	Slerp to new position	1.2s	Power2.inOut
Modal open	Fade + scale from 0.9	0.3s	Back.out
Filter hover	Background opacity 0 → 0.2	0.2s	Power1.out

📱 Responsive Breakpoints

Breakpoint	Globe Size	Layout Changes
Desktop (>1024px)	70vh, centered	Full side panels
Tablet (768-1024px)	60vh, centered	Collapsed side panels
Mobile (<768px)	50vh, full width	Bottom sheet for details, swipe rotation

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⚡ Performance Targets

- First Contentful Paint: < 1.5s
- Time to Interactive: < 3s
- Frame Rate: 60fps on mid-tier devices
- Texture Size: Max 2048x1024 (compressed)
- Bundle Size: < 500KB (Three.js tree-shaken)

🔮 Future Enhancements

- VR mode (WebXR)
- Real-time ISS tracking overlay

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- Weather data visualization on globe
- Multi-language support
- Dark/Light theme toggle
- Project collaboration graph (3D network)
- Sound design (ambient space audio)

🏁 Getting Started (Quick Setup)

```
# 1. Clone template
git clone https://github.com/yourname/3d-earth-portfolio.git
cd 3d-earth-portfolio

# 2. Install dependencies
npm install three gsap

# 3. Add your projects to assets/projects.json
# 4. Replace textures in assets/textures/
# 5. Update name/tagline in index.html

# 6. Run locally
npx vite

# 7. Deploy
npm run build
# Upload dist/ to Vercel/Netlify
```

Plan created for: Interactive 3D Earth Portfolio Reference: Rajat Sharma's portfolio design Date: July 2026

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